

Ikram Aloui

Software Engineer: Go, Kubernetes & Cloud-Native Security

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SUMMARY

Software engineer with merged pull requests in Harbor, the CNCF-graduated container registry, and its ecosystem. That code was reviewed and accepted by upstream maintainers, including a feature shipped in Harbor v2.15.0 and two security fixes backported to the release branch. Works daily in a multi-agent Claude Code workflow, defining agent tasks, reviewing agent-written code, and deciding what is accepted into production. Holds an M.Sc. in Information Systems Security.

TECHNICAL SKILLS

- Languages:** Go, Python, TypeScript/JavaScript, Java, C, SQL/PLSQL, Bash
- AI & Agent Tooling:** Claude Code, Gemini, Codex; multi-agent workflows, agent task design and evaluation, AI code review, code acceptance decisions
- Backend & Infrastructure:** REST APIs, PostgreSQL, Redis, Docker, Kubernetes, K3s, RKE2, Helm, containerd, OCI
- Security Engineering:** Cosign, Sigstore, SBOM (CycloneDX, SPDX), VEX, Trivy, SPIFFE/SPIRE, mTLS, CodeQL, threat modelling
- Cloud & CI/CD:** AWS (EKS, RDS, S3, IRSA), OpenStack, GitHub Actions, GitLab, Ansible, Hugo
- Testing & Review:** Go unit and regression testing, upstream code review, technical writing and documentation

EXPERIENCE

Software Engineer (Part-time) · 8gears AG Oct 2025 – Present
Zurich, Switzerland (Remote) · Intern until Jul 2026

Software engineering on Harbor, the CNCF-graduated container registry, and its ecosystem, working daily in a **multi-agent AI coding workflow** and owning what reaches production. Spans backend Go, security and supply-chain work, code review, and debugging across the stack, carried through upstream maintainer review to merge.

LFX Mentee, Harbor Satellite · The Linux Foundation / CNCF Mar 2026 – May 2026
Remote · Competitive paid CNCF mentorship, Term 1, 2026

- Selected from an open competitive application process, then designed and implemented a **security audit logging system** end to end in Go: event schema, a pluggable transport interface, and two transport implementations.
- Iterated the design across 23 commits and multiple review rounds, refining the transport interface boundary and testing strategy before merge.
- Authored the project's reference architecture and deployment guides.

Cybersecurity Intern · CERIST Feb 2025 – Jul 2025
Algiers, Algeria · M.Sc. thesis project

- Built a Zero Trust Kubernetes platform from scratch: Cilium/eBPF micro-segmentation, Istio mTLS, automated certificate management, and a full CI/CD chain.
- Integrated real-time observability with Prometheus, Jaeger, and Kiali plus dynamic network policies, and automated the DevSecOps chain with Harbor, Trivy, and cert-manager.

OPEN SOURCE & PROJECTS

- Cosign v3 bundle signature format**, shipped in Harbor v2.15.0 goharbor/harbor #22628
- Credential exfiltration fix in registry ping**, an access-control flaw where request parameters overrode saved server state goharbor/harbor #23463
- Process crash from an unchecked type assertion**, root-caused in an unrecovered goroutine and fixed with a recovery guard goharbor/harbor #23461

Security audit logging with syslog and OpenTelemetry, verified end to end against Wazuh and Splunk

harbor-satellite #448

harbor-scanner-devguard, a standalone Go service implementing Harbor's pluggable scanner API: crane image pulls, CycloneDX and SPDX SBOMs from a single syft pass, Cosign signing, and VEX-enriched results

personal project

EDUCATION

M.Sc., Information Systems Security · USTHB, Algiers

Sep 2023 – Jul 2025

Intrusion detection, cloud security, Zero Trust architecture, secure system administration.

B.Sc., Computer Science · USTHB, Algiers

Sep 2020 – Jun 2023

Algorithms, data structures, databases, operating systems, and programming languages.

TECHNICAL WRITING & APPLIED RESEARCH

Hardening Harbor on AWS: Zero-Static-Secret Architecture (container-registry.com engineering blog). Eliminating long-lived credentials with RDS IAM Authentication and IRSA, including the upstream code changes required.

K3s + Harbor Satellite at the Edge and **Deploying Harbor Satellite with SPIFFE/SPIRE** (satellite.container-registry.com). Reference architecture and step-by-step deployment guides.

Three technical reports (2025) on secure software supply chains, runtime behavioural detection, and service-mesh observability, each built from scratch on self-managed Kubernetes and validated adversarially rather than by assertion.

CERTIFICATIONS & LANGUAGES

Microsoft Azure Fundamentals (2024) · Ethical Hacking: Cloud Computing, Cybrary (2024) · TryHackMe Advent of Cyber (2023)

Arabic (native) · French (proficient) · English (fluent)